

Boundedness results in birational geometry

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August, 2010

Outline of the talk

- 1 Introduction
- 2 Surfaces
- 3 Higher dimensional boundedness
- 4 Generalization to log pairs

Projective varieties

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- Then $\mathcal{L} = \phi_{\mathcal{L}}^* \mathcal{O}_{\mathbb{P}^N}(1)$.
- Problem: It is hard to find natural (very ample) line bundles on X .

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- If $g = 0$ then $X \cong \mathbb{P}^1$ and $H^0(X, \omega_X^{\otimes m}) \cong 0$ for all $m > 0$.
- If $g = 1$ then X is an elliptic curve and $H^0(X, \omega_X^{\otimes m}) \cong \mathbb{C}$ for all $m > 0$.

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Theorem

Let X be a curve of genus $g \geq 2$, then the sections of $H^0(X, \omega_X^{\otimes 3})$ define an embedding of degree $6g - 6$

$$X \hookrightarrow \mathbb{P}_{\mathbb{C}}^{5g-6}.$$

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- To obtain a compact moduli space, we must allow stable curves (nodal singularities).

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- The **Kodaira dimension** of X is $\kappa(X) = \text{tr.deg.}_{\mathbb{C}} R(\omega_X) - 1$.
- We have $\kappa(X) = \max\{\dim \phi_{\omega_X^{\otimes k}}(X) \mid k > 0\}$ so that $\kappa(X) \in \{-1, 0, 1, \dots, d\}$.

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- It follows that $R(X)$ is finitely generated $\mathbb{C}$ -algebra which captures the geometry of X . In fact $X = \text{Proj} R(\omega_X)$.
- In coordinates this may be described as follows. If $r_0, \dots, r_m$ are generators of $R(\omega_X)$ and $K = \ker(\mathbb{C}[x_0, \dots, x_m] \rightarrow R(\omega_X))$, then $X \subset \mathbb{P}_{\mathbb{C}}^m$ is defined by the equations in K .

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- Thus we study varieties modulo birational equivalence.
- Given a variety X , we would like to use the canonical ring to find a distinguished representative of its birational class.

Canonical models

Theorem (Birkar-Cascini-Hacon-M^cKernan, Siu (2006))

The canonical ring

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- We say that X_{can} is the **canonical model** of X .

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- If $\kappa(X) = 0$, then X is birational to a K3, abelian, Enriques or hyperelliptic surface.
- If $\kappa(X) = 1$, then (up to birational isomorphism) there is a morphism $X \rightarrow B$ whose general fiber F is a curve of genus 1.

Birational geometry of surfaces II

- If $\kappa(X) = 2$, then by a result of Bombieri, $\phi_{\omega_X^{\otimes 5}} : X \rightarrow \mathbb{P}^N$ is a morphism whose image $X_{can} \cong \text{Proj } \bigoplus_{m \geq 0} H^0(\omega_X^{\otimes m})$ is given by contracting -1 and -2 curves (i.e. curves $E \cong \mathbb{P}^1$ with $E^2 = -1$ or $E^2 = -2$).

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- X_{can} has Du Val singularities and degree $25K_{X_{can}}^2$ in $\mathbb{P}^N$.
($K_{X_{can}}$ is the divisor class corresponding to $\omega_{X_{can}}$.)
- Therefore, if $K_{X_{can}}^2 \leq M$ then there are only finitely many families of such surfaces.

Birational geometry of surfaces III

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- Note that the moduli space of K3 surfaces has countably many irreducible components! (The problem is that $\omega_X = \mathcal{O}_X$ so one can not use multiples of the canonical line bundles to study X .)
- The moduli spaces of (canonically polarized) surfaces of general type were constructed by Gieseker.
- Canonically polarized varieties are varieties with canonical singularities such that ω_X is ample (i.e. $\omega_X^{\otimes m}$ is very ample for $m \gg 0$).

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Boundedness

The above result generalizes to higher dimensional varieties.

Theorem (Tsuji, Hacon-M^cKernan, Takayama)

For any integer $d > 0$, there is an integer $r_0 = r_0(d)$ such that if X is a projective variety of general type and dimension d then $\phi_{\omega_X^{\otimes r}}$ is birational for all $r \geq r_0$.

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Theorem

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In fact there exists a moduli space for canonically polarized varieties of dimension d and fixed K_X^d .

Consequences

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- There are examples (Iano-Fletcher) showing that if $d = 3$, $r_0 \geq 27$ and $V \leq 1/420$.

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- If $K_X^d < 1$, then $\deg \phi_{r_0}(X) \leq r_0^d \cdot K_X^d < (A + B + 1)^d$ so that X is birationally bounded.

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- One then concludes that K_X^d is bounded from below.
- To prove $r_0 \leq A + B/(K_X^d)^{1/d}$ we proceed as follows.

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- Since vanishing at x to order j imposes at most $j^d / d! + O(j^{d-1})$ conditions, there is a divisor $D_x^m \sim mK_X$ such that $\text{mult}_x(D_x^m) \cong m \cdot K_X^d$.

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- Therefore we have a section of $H^0(\omega_X^{\otimes \lambda+1})$ which doesn't vanish at x (where $\lambda = d / K_X^d$).

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- Using the MMP, we can lift these sections to get the required sections of $\omega_X^{\otimes \rho(1+\lambda)}$.

Outline of the talk

- 1 Introduction
- 2 Surfaces
- 3 Higher dimensional boundedness
- 4 Generalization to log pairs
 - Automorphisms of varieties of general type
 - Stable log Calabi-Yau fibrations
 - Stable varieties
 - Boundedness of Fanos and the ACC for log Calabi-Yau's

Varieties of log general type

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- Eg. $I = \{1 - \frac{1}{n} | n \in \mathbb{N}\}$, then $\deg(K_X + D) \geq \frac{1}{42}$.
- In fact one easily sees that the set $V = \{\deg(K_X + D) | d_i \in I\}$ satisfies the DCC.

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- In work in progress with McKernan and Xu, we prove the above conjectures when $I = \bar{I} \subset \mathbb{Q}$.

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Theorem

For any $d > 0$ there exists a constant $C = C(d) > 0$ such that if X is a d -dimensional projective variety of general type, then

$$|G| \leq C \cdot \text{vol}(\omega_X).$$

Automorphisms of varieties of general type II

- Let $f : X \rightarrow Y = X/G$, then there is a divisor $D = \sum d_i D_i$ with $d_i \in \{1 - \frac{1}{n} | n \in \mathbb{N}\}$ such that $K_X = f^*(K_Y + D)$.

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- Note that the pair (Y, D) is lc and $K_Y + D$ is big so this lower bound follows from the conjecture (but its proof is much easier).

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- For $N \geq 4$ this seems a very hard problem.

Intermediate Kodaira dimension II

- With current methods, it seems the best that we can do is to find N depending on $\dim X$ and the general fiber F . (More precisely on $b = \min\{t > 0 \mid H^0(\omega_F^{\otimes t}) \neq 0\}$ and B_m an upper bound for the middle Betti number of the canonical cover $F' \rightarrow F$ of the general fiber F .)

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- The idea is to use a result of Fujino-Mori which roughly speaking says that $R(\omega_X) \cong R(\omega_Y(D))$ where $Y = \text{Proj} R(\omega_X)$ and (Y, D) is a klt pair of log general type. The coefficients of D belong to a DCC set $I = \bar{I} \subset \mathbb{Q}$ depending on F .

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- Once again the key issue is to show that these varieties (or better the pairs (X_i, D_i)) are bounded.

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- It has many important applications. Eg. to the ACC for mld's, the ACC for log pairs and the termination of flips.
- We hope that some of these applications can be deduced from results on the boundedness of pairs of general type.

ACC for lct's

- Let $(X, D = \sum d_i D_i)$ be a klt pair and $G = \sum g_i G_i$ an effective $\mathbb{Q}$ -divisor. Then the log canonical threshold is

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- This conjecture seems to follow (work in progress) from our results on the boundedness of varieties of log general type.

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- We may assume that the center of E_i is a point in X_i , so that

$$K_{E_i} + \Delta_i = f_i^*(K_{X_i} + D_i + \tau_i G_i)|_{E_i} \equiv 0.$$

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$$(f'_i)^*(K_{X_i} + D_i + \tau_i G_i) = K_{X'_i} + (f_i^{-1})_*(D_i + \tau_i G_i) + E_i.$$
- We may assume that the center of E_i is a point in X_i , so that $K_{E_i} + \Delta_i = f_i^*(K_{X_i} + D_i + \tau_i G_i)|_{E_i} \equiv 0$.
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ACC for lct's II

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- But then $K_{E_i} + N(K_{E_i} + \Delta'_i)$ is big and as $K_{E_i} + N(K_{E_i} + \Delta'_i) \leq (N+1)(K_{E_i} + \Delta_i)$ for $i \gg 0$, we have the required contradiction.